

Cost Analysis - Property RAG System

Date: October 2025 **Project:** Property Data RAG System **Purpose:** Simplyphi Recruitment Submission

Executive Summary

This document provides a comprehensive cost analysis for the Property RAG System, covering development, deployment, scaling, and operational costs across different scenarios.

Key Findings: - **Current Setup:** \$0/month (100% free tier usage) - **Small Scale (1000 queries/day):** ~\$15-25/month - **Medium Scale (10,000 queries/day):** ~\$150-200/month - **Enterprise Scale (100,000 queries/day):** ~\$1,500-2,000/month

1. Current Implementation Costs

1.1 Development Stack (FREE)

Component	Technology	Cost	Notes
Vector Database	ChromaDB (Local)	\$0	Self-hosted, persistent storage
Embedding Model	Sentence-Transformers	\$0	Runs locally, no API calls
LLM	Google Gemini Pro (Free Tier)	\$0	15 req/min, 1500 req/day limit
Backend	FastAPI	\$0	Open source framework
Frontend	Streamlit	\$0	Open source framework
Hosting (Dev)	Localhost	\$0	Development environment

Total Development Cost: \$0/month

1.2 Infrastructure Requirements

Minimal Setup: - RAM: 4GB minimum (8GB recommended) - Storage: 1GB for code + 500MB for vector DB - CPU: 2 cores minimum - Bandwidth: Minimal (local processing)

Cost for Self-Hosted: \$0 (using existing hardware)

2. Production Deployment Costs

2.1 Small Scale Deployment (1,000 queries/day)

Infrastructure:	Component	Provider	Specifications	Cost/Month
Server	Digital Ocean Droplet		2 vCPU, 4GB RAM	\$24
	Alternative	AWS Lightsail	2 vCPU, 4GB RAM	\$20
Alternative	Heroku Basic		512MB RAM	\$7

Services:	Service	Provider	Usage	Cost/Month
LLM	Google Gemini Pro		1,000 queries/day	\$0 (free tier)
	Alternative LLM	OpenAI GPT-3.5	1,000 queries × \$0.001	\$30
Vector DB	ChromaDB (self-hosted)		Included in server	\$0
	Alternative Vector DB	Pinecone Starter	100K vectors	\$0 (free tier)

Total Small Scale (Gemini): \$20-24/month **Total Small Scale (OpenAI):** \$50-54/month

2.2 Medium Scale Deployment (10,000 queries/day)

Infrastructure: | Component | Provider | Specifications | Cost/Month |
|-----|-----|-----|-----| | **Server** | AWS EC2 t3.medium | 2 vCPU, 4GB RAM | \$30 | | **Load Balancer** | AWS ALB | Basic | \$16 | | **Storage** | AWS EBS | 50GB SSD | \$5 |

Services: | Service | Usage | Cost/Month | |-----|-----|-----| | **LLM (Gemini Pro)** | 10,000 queries/day × \$0.001 | \$300 | | **LLM (GPT-3.5 Turbo)** | 10,000 queries/day × \$0.001 | \$300 | | **Embeddings** | Self-hosted Sentence-Transformers | \$0 | | **Vector DB** | Pinecone Standard (1M vectors) | \$70 | | **Alternative** | Weaviate Cloud | 10M vectors | \$25 | | **Monitoring** | Datadog Basic | 5 hosts | \$15 |

Scenarios:

Option A - Gemini + Self-hosted ChromaDB: - Infrastructure: \$51/month - LLM: \$300/month - **Total: ~\$351/month**

Option B - Gemini + Managed Vector DB: - Infrastructure: \$51/month - LLM: \$300/month - Vector DB: \$70/month - **Total: ~\$421/month**

Option C - OpenAI + Pinecone: - Infrastructure: \$51/month - LLM: \$300/month - Vector DB: \$70/month - **Total: ~\$421/month**

2.3 Enterprise Scale (100,000 queries/day)

Infrastructure: | Component | Specifications | Cost/Month | |-----|-----|-----| | **Kubernetes Cluster** | 3 nodes × t3.large (2 vCPU, 8GB each) | \$150 | | **Load Balancer** | AWS ALB with SSL | \$25 | | **Storage** | 200GB SSD | \$20 | | **CDN** | CloudFlare Pro | \$20 | | **Monitoring & Logging** | Datadog Pro | \$50 |

Services: | Service | Usage | Cost/Month | |-----|-----|-----| | **LLM (Gemini Pro 1.5)** | 100K queries/day @ \$0.001/query | \$3,000 | | **Alternative: OpenAI** | 100K queries/day @ \$0.001/query | \$3,000 | | **Alternative: Claude** | 100K queries/day @ \$0.002/query | \$6,000 | | **Vector DB (Pinecone)** | 10M vectors, Standard | \$100 | | **Alternative: Weaviate** | 100M vectors | \$200 | | **Caching (Redis)** | AWS ElastiCache | \$50 |

Total Enterprise Scale: - Infrastructure: \$265/month - Services (Gemini + Pinecone): \$3,150/month - **Total: ~\$3,415/month**

3. Cost Optimization Strategies

3.1 LLM Cost Reduction

Strategy 1: Hybrid Approach - Use Gemini free tier for first 1,500 queries/day - Fallback to paid tier only when exceeded - **Savings:** ~\$45/month for small scale

Strategy 2: Query Caching - Cache common queries and responses - Estimated cache hit rate: 30-40% - **Savings:** 30-40% reduction in LLM calls

Strategy 3: Local LLM for Simple Queries - Use Ollama (free, local) for simple queries - Use cloud LLM only for complex queries - **Savings:** 50-60% reduction in API costs

3.2 Infrastructure Optimization

Strategy 1: Spot Instances - Use AWS Spot instances (70% discount) - **Savings:** ~\$100/month for enterprise scale

Strategy 2: Reserved Instances - 1-year commitment: 30% discount - 3-year commitment: 50% discount - **Savings:** ~\$150/month for enterprise scale

Strategy 3: Serverless Architecture - AWS Lambda for API (pay per use) - Only pay when queries are processed - **Ideal for:** Variable/bursty traffic

3.3 Vector Database Optimization

Strategy 1: Compression - Use quantization for embeddings - Reduce vector dimensions (384 → 256) - **Savings:** 33% storage cost reduction

Strategy 2: Tiered Storage - Hot data: Fast SSD storage - Cold data: S3 archival - **Savings:** ~40% for historical data

4. Scaling Cost Projections

4.1 Cost Per Query Analysis

Scale Queries/Day Monthly Queries Infrastructure LLM Vector DB Total/Month Cost/Query												
----- ----- ----- ----- ----- ----- ----- -----												
Dev 100 3,000 \$0 \$0												
\$0 \$0 \$0 Small 1,000 30,000 \$24 \$0 \$0 \$24 \$0.0008 Medium 10,000 300,000 \$51												
\$300 \$70 \$421 \$0.0014 Large 50,000 1,500,000 \$150 \$1,500 \$100 \$1,750 \$0.0012												
Enterprise 100,000 3,000,000 \$265 \$3,000 \$100 \$3,365 \$0.0011												

Key Insight: Cost per query decreases with scale due to infrastructure efficiency.

4.2 Break-Even Analysis

Free Tier Limits: - Gemini: 1,500 queries/day free - After limit: \$0.001 per query

When to Upgrade: - **Stay Free:** < 1,500 queries/day - **Basic Paid:** 1,500 - 10,000 queries/day - **Managed Services:** > 10,000 queries/day

5. Alternative Architectures & Costs

5.1 Fully Serverless (AWS)

Component Service Cost Model ----- ----- -----	Compute AWS Lambda \$0.20 per 1M requests + \$0.0000166667 per GB-second
	Vector DB Pinecone Serverless Pay per query (\$0.0001/query)
	LLM Bedrock Claude \$0.003 per 1K input tokens
	Storage S3 \$0.023 per GB
Estimated Cost (10K queries/day): - Lambda: ~\$15/month - Pinecone: ~\$30/month - LLM: ~\$300/month	
- Total: ~\$345/month	

5.2 Fully Open Source (Self-Hosted)

Component	Technology	Cost	----- ----- -----	Server	Own hardware/VPS	\$30-50/month	
Vector DB	ChromaDB/Milvus	\$0		Embeddings	Sentence-Transformers	\$0	
LLM	Ollama (Llama 3 8B)	\$0		Monitoring	Prometheus + Grafana	\$0	

Total: \$30-50/month (ANY scale)

Trade-offs: - Lower quality LLM responses - Higher latency - More maintenance overhead - Full data privacy

6. Cost Comparison: Cloud Providers

6.1 Vector Database Comparison

Provider	Free Tier	Paid Tier	Notes
Pinecone	100K vectors \$70/month for 1M	Managed, easy setup	
Weaviate Cloud	100K vectors \$25/month for 10M	Open source based	
Qdrant Cloud	None	\$0.40 per GB/month	Cost-effective at scale
ChromaDB	Unlimited (self-hosted)	Free	Requires own infrastructure

6.2 LLM Comparison

Provider	Model	Input Cost	Output Cost	Context
Google Gemini	Pro	Free (limited)	then \$0.0005/1K	\$0.0015/1K 30K tokens
OpenAI	GPT-3.5 Turbo	\$0.0005/1K	\$0.0015/1K	16K tokens
Anthropic	Claude 3 Haiku	\$0.00025/1K	\$0.00125/1K	200K tokens
Meta	Llama 3 (Ollama)	Free (self-hosted)	Free	8K tokens

7. ROI & Business Case

7.1 Value Proposition

For 10,000 queries/day @ \$421/month: - Cost per query: \$0.0014 - If each query saves 5 minutes of manual search: 833 hours/month saved - At \$20/hour labor cost: **\$16,660 value generated - ROI: 3,855%**

7.2 Pricing Strategy for SaaS

Suggested Pricing Tiers:

Tier	Queries/Day	Cost to Serve	Suggested Price	Margin
Free	10	\$0	\$0	-
Starter	100	\$0.10	\$10/month	99%
Professional	1,000	\$1	\$50/month	98%
Business	10,000	\$14	\$200/month	93%
Enterprise	Custom	Variable	Custom	80-90%

8. Cost Monitoring & Alerts

8.1 Recommended Alerts

Metric	Threshold	Action
LLM API Cost	> \$100/day	Switch to cached responses
Query Volume	> 10K/day	Upgrade infrastructure
Error Rate	> 5%	Investigate issues
Response Time	> 3s average	Scale up resources

8.2 Cost Tracking Tools

AWS Cost Explorer: Track AWS spending **Pinecone Dashboard:** Monitor vector DB usage **Custom Analytics:** Track query patterns and optimize

9. Recommendations

9.1 For Current Scale (Development)

■ **Use Current Stack:** - ChromaDB (self-hosted): \$0 - Gemini Free Tier: \$0 - Local hosting: \$0 - **Total: \$0/month**

9.2 For Production (1-10K queries/day)

■ **Recommended:** - AWS Lightsail/DO Droplet: \$20-30/month - Gemini Pro (paid): \$300/month - Self-hosted ChromaDB: \$0 - **Total: ~\$320-330/month**

9.3 For Enterprise Scale

■ **Recommended:** - Kubernetes cluster: \$150/month - Gemini Pro with caching: \$2,000/month (33% savings) - Weaviate Cloud: \$200/month - Redis caching: \$50/month - **Total: ~\$2,400/month**

10. Conclusion

The Property RAG System is designed to be **cost-effective at all scales**:

Development: \$0/month (100% free) **Small Production:** ~\$25/month **Medium Scale:** ~\$350-450/month

Enterprise: ~\$2,400-3,400/month

Key Cost Drivers: - LLM API calls (70-80% of costs) - Vector database (10-15% of costs) - Infrastructure (10-15% of costs)

Optimization Potential: - Query caching: 30-40% savings - Hybrid LLM approach: 50% savings - Spot instances: 30-50% savings - **Total potential savings: 60-70%**

Final Verdict: The system provides exceptional value with costs scaling linearly with usage while maintaining high performance and accuracy.

Prepared by: Krishna **For:** Simplyphi Recruitment Process **Last Updated:** October 2025